Stat Mech Hw#1

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Prob 1.)



(No exch of heat, mass, or work.)

- heat leaves the reservoir.

Objective:

Prove: $\Delta S \geq \frac{Q}{T_{RES}}$

Given:

isolated system + reservoir:

- $\Delta Q = 0$, $\Delta m = 0$, $\Delta W = 0$.

1st Law of thermo:

- $\Delta E = \Delta Q + \Delta W$

2nd Law of thermo, Chandler:

$\Delta S_{Tot} \geq 0$

entropy: $S = \frac{Q}{T}$

a.) • $\Delta U_{total} = \Delta U_{reservoir} + \Delta U_{system}$

• $\Delta U_{Total} = 0$

• Isolated, • (no heat, mass, or work exchange.)

• $\Delta S_{reservoir} = \frac{-Q}{T_{res}}$

• $\Delta S_{system} = \frac{Q}{T}$

• T_{res} → temperature of reservoir.

• $\Delta S_{total} = \frac{Q}{T} - \frac{Q}{T_{res}} = \Delta S_{res} + \Delta S_{sys}$

• (Chandler's: $\Delta S_{total} \geq 0$)

• $\frac{Q}{T} - \frac{Q}{T_{res}} \geq 0$

• $\frac{Q}{T} \geq \frac{Q}{T_{res}}$ $\left(\bullet \frac{Q}{T} = \Delta S_{sys}\right)$

• $\boxed{\Delta S_{sys} \geq \frac{Q}{T_{res}}}$

b) 1st Law of therm:

- $\Delta U = = \Delta Q - \Delta W$
  - Isothermal (constant temp)
  - Isochoric (constant volume)
- $W \rightarrow PV$ (constant volume $\Delta W = 0$)
- $\Delta U = \Delta Q$
  - $A = U - TS$
  - $\Delta A = \Delta U - T\Delta S$ ($\Delta T = 0$)
- $\Delta A = \Delta U$
- $\Delta A = Q$, The system does positive work if it does
- $W_{rev} = -Q$ (Reversible process)
- $W_{rev} = -\Delta A$

Prob 2.)

a.) • Idea gas law the gas:

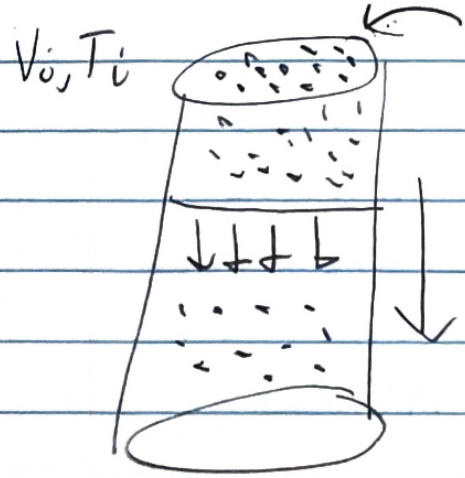
1.) $pV = nRT$

2.) $\left(\frac{\partial E}{\partial V}\right)_{n,T} = 0$

3.) Reversible adiabatic process:
- $pV^{\gamma} = \text{const}$
- $\gamma = C_p/C_V$

Process # 1:

- $n \rightarrow$ number of moles of gas.
- $V_i \rightarrow$ initial volume.
- $T_i \rightarrow$ initial temperature.
- $V_f \rightarrow$ Final volume after expansion.



• gases expanding would be negative work.

• $dE = \left(\frac{\partial E}{\partial V}\right)_{nT} dV + \left(\frac{\partial E}{\partial n}\right)_{VT} dn + \left(\frac{\partial E}{\partial T}\right)_{V,n} \partial T$

$= 0$

Heat: process is adiabatic

$\boxed{Q = 0}$

Work (W):

• Work is negative because of gas expansion.

$P = 0$ — (zero pressure)

$\boxed{W = 0}$

Energy:

$\Delta E = \Delta Q - \Delta W$

$\Delta Q = 0$

$\Delta W = 0$

$\boxed{\Delta E = 0}$

Constant temperature:

$\Delta E = n C_V \Delta T$

$\Delta E = 0, \quad \boxed{\Delta T = 0}$

Entropy

$$dE = TdS - PdV$$

- $dU = 0$
- $TdS = PdV$
- $P = \frac{nRT}{V}$
- ~~$TdS = \frac{nRT}{V} dV$~~

$V_i \leq V \leq V_f$

- $\int_{S_i}^{S_f} dS = nR \int_{V_i}^{V_f} \frac{dV}{V}$
- $\boxed{\Delta S = nR \ln\left(\frac{V_f}{V_i}\right)}$

- Since we're in vacuum we can ignore chemical potential term.

Process II :

- Reversible Isothermal expansion
- $\boxed{\Delta T = 0}$ Temperature
- Δ Internal energy:

$$\boxed{\Delta E = nC_V \Delta T = 0}$$

$(\Delta T = 0)$

- Work & Heat:

$dE = dQ - dW$, $dE = 0$

dQ